29-04-2026

Agenda:

Feature Engineering - 4

1) visualization •
2) IQR •
3) z-score • → normal distribution ≠ → z-score = 3 •
   → modified zscore •
4) winsorization •

z-score :

$$Z = \frac{x - \mu}{\sigma}$$

μ (mean)
σ (std dev) → cutoff ± 3

modified z-score :—

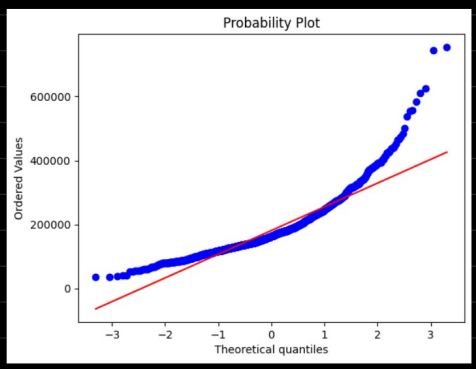
skewed data

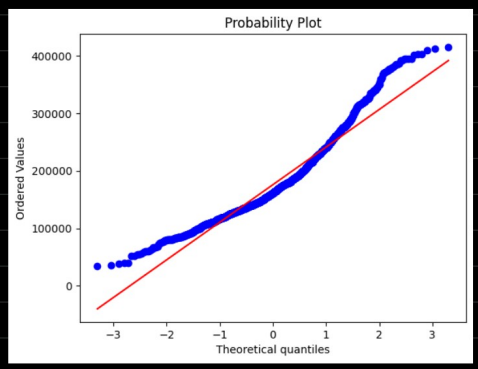
$$M = \frac{0.6745 \times (x - \tilde{x})}{MAD}$$

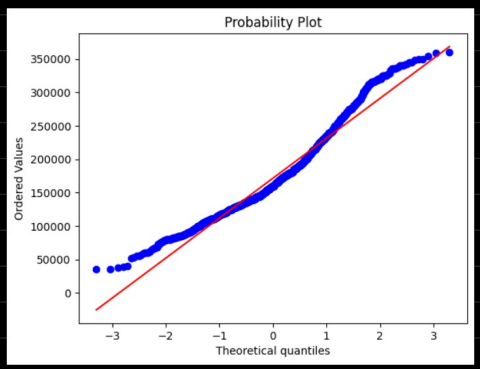
$\tilde{x}$ (median)

MAD → median absolute deviation
↓
median of deviation from the median.

→ cutoff ± 3.5







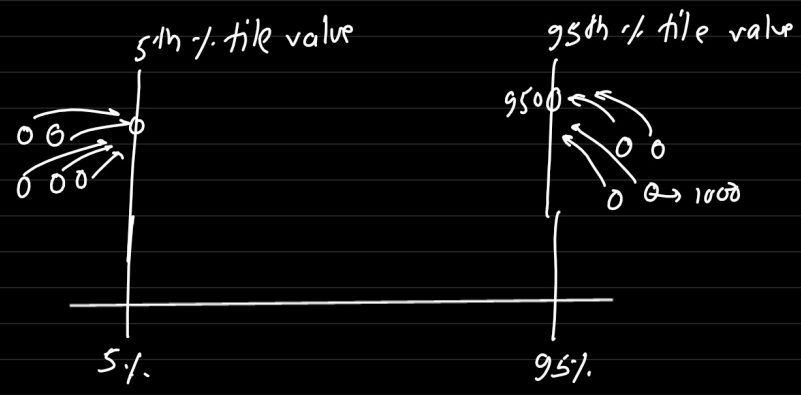
with outliers

without outliers (zscore)

without outliers (mod zscore)

fold = 0.05

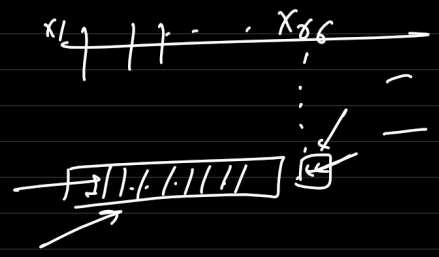
tail = both



capping extreme values at some percentile

→ row count doesn't change

→ reduces range of extremes



Feature Scaling

↓ x humans

ML

why feature scaling is needed ?

* Many ML Algo (Linear Regression, SVM, kNN, NN) are distance based or assume feature are on comparable scale

* Age ∈ [0, 100]
  salary ∈ [0, 100000]

* Scaling ensures all feature contribute equally.

x_1 ↓ Age	x_2 ↓ salary (k)	↓ y salary_estimated_5_years (k)
24	30,000	35,000
26	36,000	40,000
30	40,000	46,000
40	80,000	86,000
ML → 38 m1	42,000	52,000

• m1

• m2

$x_1 m_1 + x_2 m_2 = 52000$

$\downarrow$ 2.6 $\quad$ 1.2

$38 \times 2.6 = 98.8$ $\quad$ $42{,}000 \times 1.2 = 50400$

$= 50400 + 98.8$

$= 50498.8$

$= 52000 - 50498.8$

$= \boxed{1501.2}$

38 ← m1 $\quad$ m2 → 42,000

80.8 ↕ $\quad$ 8400 ↕

adjust

98.8 ← $\quad$ → 50400

formulas for each technique:

(1) Min max scaling (Normalization)

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}} \longrightarrow [0,1]$$

Age → 26 ⟷ 81

min → 0 , max → 1

$$\frac{26-26}{81-26} = \frac{0}{55} = 0$$

$$\frac{81-26}{55} = \frac{55}{55} = 1$$

→ sensitive to outliers

(2) Standardization (z score scaling)

$$x' = \frac{x - \mu}{\sigma}$$

mean = 0, std = 1

Handles outliers better than min-max scaling

(3) Robust scaler (Median & IQR based)

$$x' = \frac{x - \text{median}(x)}{IQR(x)}$$

Robust to outliers

(4) log Transformation

$$x' = \log(1 + x)$$

→ reduces right skewness

Note: (For left skewness, we use:)

→ square root / cube root

→ Box-Cox (positive data)

→ Yeo - Johnson (works with negative too)